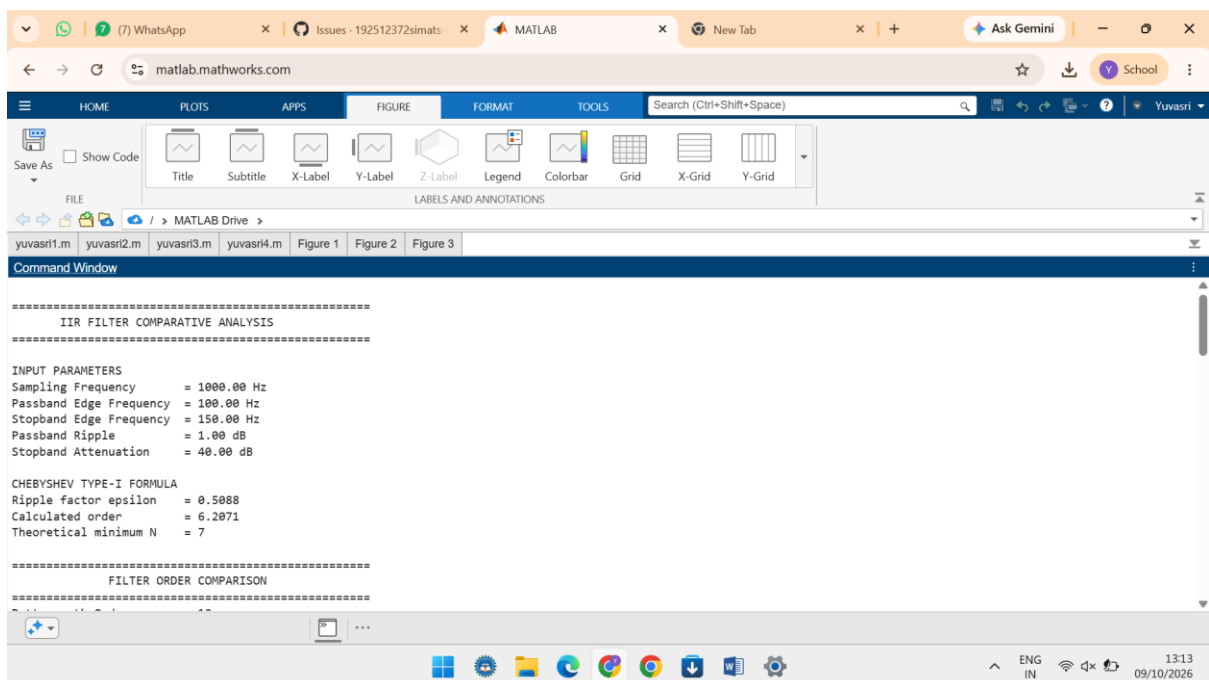
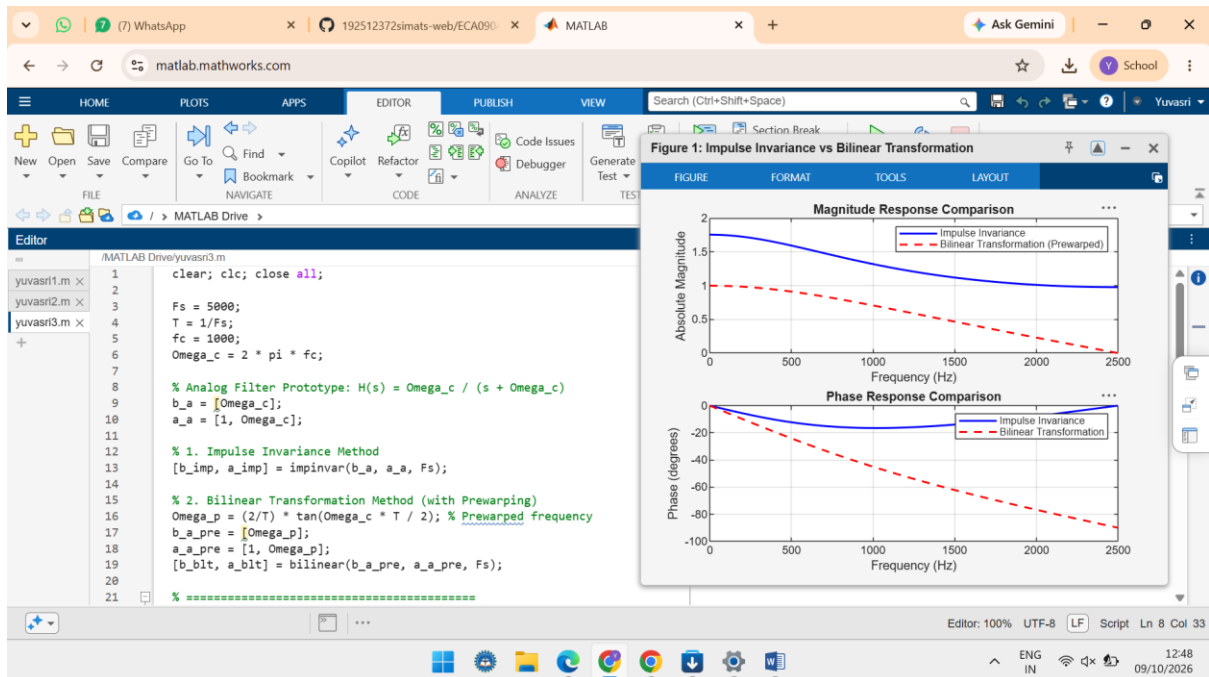


3. Design of IIR Filters Using Impulse Invariance and Bilinear Transformation Methods Using MATLAB



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HOME PLOTS APPS **FIGURE** FORMAT TOOLS Search (Ctrl+Shift+Space)

Save As Show Code

FILE

Labels and Annotations: Title Subtitle X-Label Y-Label Z-Label Legend Colorbar Grid X-Grid Y-Grid

yuvasri1.m yuvasri2.m yuvasri3.m yuvasri4.m Figure 1 Figure 2 Figure 3

Command Window

```

Butterworth Order      = 12
Chebyshev Type-I Order = 6
Elliptic Order         = 4

=====
BUTTERWORTH COEFFICIENTS
=====
Numerator b =
1.0e-03 *

    0.0002    0.0027    0.0146    0.0488    0.1098    0.1757    0.2050    0.1757    0.1098    0.0488    0.0146    0.0027    0.0002

Denominator a =
    1.0000   -6.8844   22.4341  -45.5100   63.7725  -64.8501   48.9643  -27.6092   11.5218   -3.4662    0.7128   -0.0899    0.0053

=====
CHEBYSHEV TYPE-I COEFFICIENTS
=====
Numerator b =

```

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Save As Show Code

FILE

Labels and Annotations: Title Subtitle X-Label Y-Label Z-Label Legend Colorbar Grid X-Grid Y-Grid

yuvasri1.m yuvasri2.m yuvasri3.m yuvasri4.m Figure 1 Figure 2 Figure 3

Command Window

```

    0.0464    0.2782    0.6956    0.9274    0.6956    0.2782    0.0464

Denominator a =
    1.0000   -4.8694   10.3807  -12.3367    8.5969   -3.3268    0.5586

=====
ELLIPTIC COEFFICIENTS
=====
Numerator b =
    0.0197   -0.0171    0.0333   -0.0171    0.0197

Denominator a =
    1.0000   -3.0330    3.8118   -2.2911    0.5554

=====
FILTER PERFORMANCE
=====

```

13:14 09/10/2026

Browser tabs: (7) WhatsApp, Issues · 192512372simats, MATLAB, New Tab, Ask Gemini, School

Address bar: matlab.mathworks.com

Navigation: HOME, PLOTS, APPS, FIGURE, FORMAT, TOOLS

Search: Search (Ctrl+Shift+Space)

Save As: ☐ Show Code

FILE: yuvasri1.m, yuvasri2.m, yuvasri3.m, yuvasri4.m, Figure 1, Figure 2, Figure 3

COMMAND WINDOW

```
PASSBAND RIPPLE
Butterworth      = 0.8033 dB
Chebyshev Type-I = 1.0000 dB
Elliptic         = 1.0000 dB

STOPBAND ATTENUATION
Butterworth      = 40.0198 dB
Chebyshev Type-I = 41.3365 dB
Elliptic         = 40.0000 dB

CHEBYSHEV RIPPLE DEVIATION
Specified Ripple = 1.00 dB
Measured Ripple  = 1.0000 dB
Percentage Deviation = 0.0000 %

TRANSITION CHARACTERISTICS
Passband Edge    = 100.00 Hz
Stopband Edge    = 150.00 Hz
Transition Band  = 50.00 Hz
>>
```

Windows Taskbar: 13:14, 09/10/2026